

$$\frac{dy}{dt} = k_1 (x_0 - y_{\infty}) - k_{-1} (1 - x_0 + y_{\infty}) = 0$$

$$x_0 (k_1 + k_{-1}) - k_1 y_{\infty} - k_{-1} (1 + y_{\infty}) = 0$$

$$x_0 = \frac{k_1 y_{\infty} + k_{-1} (1 + y_{\infty})}{(k_1 + k_{-1})}$$

$$\begin{aligned} \frac{dy}{dt} &= (k_1 + k_{-1}) x_0 - k_1 y - k_{-1} (1 + y) \\ &= k_1 y_{\infty} + k_{-1} (1 + y_{\infty}) - k_1 y - k_{-1} (1 + y) \end{aligned}$$

$$= k_1 (y_{\infty} - y) + k_{-1} (1 + y_{\infty} - 1 - y)$$

$$\frac{dy}{dt} = (k_1 + k_{-1}) (y_{\infty} - y)$$